

ARYAN THAKAR

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Education

Nirma University, Institute of Technology, Ahmedabad, India
B.Tech, Electronics & Instrumentation Engineering

Sep 2022 – June 2026
CGPA – 7.68

Technical Skills

Embedded Systems: STM32, ESP32, RPi, Kendryte K210, PIC18F, Arduino

Programming: C, C++, Python, Assembly, MicroPython, Shell

Protocols: UART, I2C, SPI, RS485, I2S, Ethernet

Tools: STM32CubeIDE, MaixPy, MPLAB, MATLAB, Simulink, KiCad, VS Code, ST-Link

Relevant Experiences

Junior Embedded Engineer Intern | STEMbotix PVT. LTD.

Dec 2025 – April 2026

- Developed and delivered a production-ready AI-vision expansion board (Kendryte K210) for a STEM robotics kit, enabling real-time face, object, and audio detection.
- Solved a critical flash constraint by modifying firmware build config (Makefile) to strip unused MaixPy modules, fitting 1.5MB firmware + 3.69MB trained model within 16MB flash.
- Configured UART, SPI, and I2S peripherals for camera, microphone, LCD, DAC, and amplifier; board fully handed off and deployed by end of internship.

Summer Intern | iCreate (International Centre for Entrepreneurship & Technology)

May 2025 – July 2025

- Engineered a two-phase onboard controller for an underwater robot, iterating from RPi 4B to a custom STM32F411-based solution based on real-world testing outcomes.
- Phase 1 (RPi/BlueOS): Modified BlueOS to transmit H.264 video packets over RS485 alongside IMU and pressure telemetry; designed a custom framing protocol (CAM/IMU/BNO frame headers) to multiplex all streams on a single RS485 line.
- Phase 2 (STM32): Developed bare-metal C and Assembly firmware for STM32F411 with driver-level Timers, PWM, UART, and RS485; built Python GUI from scratch for real-time demultiplexing, live telemetry, and manual control.

Projects

Sensor Fusion for Real-Time Localisation | Academic Research

Aug 2025 – Oct 2025

- Developing fusion pipeline for a wheeled robot combining encoder odometry and BNO055 IMU; explored EKF, implemented custom weighted-averaging baseline with EKF integration ongoing.
- Built C++/Python toolchain for automated sensor testing, data logging, and live state visualisation.

Autonomous Hexapod Robot | RoboFest 4.0 – GUJCOST

March 2024 – Jan 2025

- Won Rs. 2,50,000 staged grant across 3 competitive phases to design and build an 18-DOF hexapod (3 DOF $\times$ 6 legs).
- Stage 1: ESP8266 AP-mode WiFi with mobile webpage for real-time gait control; Stage 2: upgraded to RPi 4B with live camera feed, IMU orientation monitoring, and 2D LiDAR point cloud visualisation streamed over network via Matplotlib.

PD Control on PIC Microcontroller | Academic Project

Feb 2023 – April 2023

- Implemented PD control algorithm on PIC18F series to regulate output against a configurable setpoint; validated response using oscilloscope measurements.

Publications

- Thakar, A., et al. "A Sustainable Approach for Plastic Waste Management and Recycling." *Recent Advances in Materials and Manufacturing Science (Springer Nature)*, 2025

Achievements

- Runners-up, Secure Meters' Vikram Awards; Runners-up, Open Automation Challenge 2024

Relevant Coursework

Control Theory, Signals & Systems, Fundamentals of Robotics, Robotic Control Systems, Industrial Controls, Embedded System Design (STM32), Fuzzy Control Theory, Data Communication